CVR COLLEGE OF ENGINEERING

(An UGC Autonomous Institution with NAAC 'A' Grade,Affiliated to JNTUH)
Vastunagar, Mangalpalli (V), Ibrahimpatan (M), R.R.District, Telangana
Ph.No:91-8414 – 661601, 661675

DEPARTMENT OF CSE (DATA SCIENCE)

COURSE OUTCOMES (COs)
OF
B. TECH. **CSE (DATA SCIENCE)**

R18 REGULATIONS

I Year I Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
68101	**English**	CO1	Write coherent, unified, and complete sentences.
		CO2	Identify word meaning and know the use of familiar lexical items.
		CO3	Understand explicit and implicit information and draw inferences for the given task.
		CO4	Communicate according to place, relation and medium.
		CO5	know, emphasize, conceptualize, comprehend, apply, synthesize, and evaluate the given text, and other authentic
68104	**Applied Physics**	CO1	The concepts involving the physics of lasers. Lasing action, construction and working of HeNe laser, semiconductor laser and propagation of light through optical fibers.
		CO2	Schrodinger wave equation and its application, free electron models, formation of bands in solids and electron occupation in bands.
		CO3	Estimation of charge carrier concentration in semiconductors and understand the formation of PN junction, construction and characteristics of different diodes like rectifying, Zener & Tunnel diodes.
		CO4	Transistor current components, characteristics of CB,CE and CC configurations, also understand the construction, working and characteristics of JFET & MOSFET.
		CO5	The principles of nanotechnology, types of Nano materials, synthesis: Top-down and bottom-up methods, characterization: XRD, SEM & TEM.
68106	**Mathematics for Computing**	CO1	Characterize the functions.
		CO2	Evaluate surface areas and volumes of solids of revolution, Apply Mean value theorems in relevant engineering domains.

		CO3	Determine the convergence/divergence of a given infinite series.
		CO4	Find the extreme of a multi-variate function with or without constraints.
		CO5	Solve real-world problems by evolving difference equations
65101	**Problem Solving through C**	CO1	Ability to understand programming concepts and analyze a problem, design a solution and develop an algorithm to solve it.
		CO2	Ability to modularize a problem and implement the solution using basic programming concepts, control statements and functions.
		CO3	Ability to evaluate the use of macros and implement solutions to complex problems using recursion and homogeneous data types.
		CO4	Ability to implement pointers for problems of relevance and use different dynamic memory allocation methods.
		CO5	Design and implement appropriate user defined structures to a given problem definition and apply various functions for processing files.
63102	**Engineering Drawing**	CO1	Know the standard conventions, design scale for drawing engineering components and draw geometrical constructions.
		CO2	Apply fundamentals of theory of projections, and draw orthographic projections of points and lines in any position.
		CO3	Construct orthographic projections of simple planes and regular solids in any position.
		CO4	Draw sectional views and developments of various basic 3D objects.
		CO5	Construct isometric views and construct multi view drawings of simple and complex 3D objects.
68131	**English language and Communication Skills Lab– I**	CO1	Emerge as good speakers and listeners.
		CO2	Develop critical and analytical thinking.
		CO3	Write effectively.
		CO4	Develop effective presentation skills using the multimedia tools.
		CO5	Neutralize mother tongue influence on their English and make them proficient speakers.
68134	**Applied Physics Lab**	CO1	Get an understanding of errors and their estimation in determination of physical measurements.
		CO2	Get an understanding of the laws of physics associated with the experiments.
		CO3	Would develop skills in handling various kinds of laboratory instruments.
		CO4	Get awareness of the magnitudes of the different physical parameters and learn how to present the observations and result at the end of an experiment.

		CO5	Get an understanding of the physical concepts involved in the experiment sent.
65131	**Computer Programming Lab**	CO1	Familiarity of programming environment in Linux operating system and to translate given algorithms to a working and correct program.
		CO2	Ability to interpret syntax errors as reported by the compilers and to be able to identify and correct logical errors encountered at run time using debuggers like GDB.
		CO3	Ability to write iterative as well as recursive programs.
		CO4	Ability to represent data in arrays, pointers, strings and structures and manipulate them through a program and use them in defining self-referential structures or structures or designing a user defined data type.
		CO5	Ability to implement file-processing functions and be able to store, retrieve and process data in text and binary formats.
63131	**Engineering Workshop**	CO1	Acquire skills of basic engineering trades like Carpentry, Tin smithy
		CO2	Demonstrate an understanding of and comply with workshop safety regulations.
		CO3	Identify and use marking out tools, measuring equipment and to work to prescribed tolerances
		CO4	Apply the knowledge of the above trades in their day–today activities.
		CO5	Select appropriate equipment and consumables for required applications

I Year II Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
68155	**Engineering Chemistry**	CO1	Rationalize periodic properties such as ionization potential, electron affinity, oxidation states and electronegativity.
		CO2	Understanding the importance of EMF, corrosion and treatment of water.
		CO3	List major chemical reactions that are used in the synthesis of molecules.
		CO4	Analyze microscopic chemistry in terms of atomic and molecular orbitals and intermolecular forces.
		CO5	Would develop ability to handle situations involving problems associated with chemical substances in engineering situations.
68156	**Environmental Science**	CO1	Define the concepts of Ecosystem and emphasize the importance of biodiversity and its conservation.

		CO2	Gain knowledge on natural resources and advantages and disadvantages on renewable energy sources and technologies.
		CO3	Develop awareness on pollution control technologies and global atmospheric changes.
		CO4	Emphasize the importance of Environmental impact assessment and green technologies.
		CO5	Understand about Environmental legislation and the concept of Sustainable development.
68158	**Applied Liner Algebra**	CO1	Deal with high dimensional data and evaluate the rank of matrices.
		CO2	Apply orthogonal transforms on large-scale data.
		CO3	Find rank of a matrix and solve a linear system of equations. Evaluate Eigen values, Eigen vectors and find the model matrix under a linear transformation.
		CO4	Decompose the matrix to optimize the computational complexity.
		CO5	Solve real-world engineering problem using least square solutions.
65151	**Data Structures through C**	CO1	Understand basic concepts, Design and implement linear data structures such as linked lists, stacks, queues by using C as the programming language using static or dynamic implementations.
		CO2	Able to understand and analyze, differentiate and implement elementary algorithms: sorting, searching and hashing and will also be able to compare and contrast algorithms with respect to time and space complexity.
		CO3	Able to implement nonlinear data structures like trees and graphs and apply appropriate data structures to designing solutions to real world complex problems.
		CO4	Demonstrate sound understanding of graph traversals and ability to implement various algorithms on graphs and interpret the results.
		CO5	Ability to implement hashing techniques for storing and searching efficiently.
69151	**Python Programming**	CO1	Know the importance of object oriented concepts and do basic Python programming.
		CO2	Write programs using various control structures, function and modules.
		CO3	Apply string operations to develop various text based applications.
		CO4	Design and develop solutions to real world problems using available data structures in Python language.

		CO5	Use of files for data storage.
68181	**English language and Communication Skills Lab – II**	CO1	Evolve as effective communicators and will develop narrative skills
		CO2	Emerge as decision makers and autonomous learners
		CO3	Develop critical and analytical kills
		CO4	Gather ideas and information, and organize them coherently.
		CO5	Develop leadership and team building skills.
65182	**Python Programming and Computational Methods Lab**	CO1	Design, code and test Python programs.
		CO2	Understand and use of functions, modules and strings in Python programs.
		CO3	Analyze problems and use Python data structures such as lists, tuples and dictionaries to develop Python programs for problems in Computational Methods.
		CO4	Apply object-oriented concepts to solve problems.
		CO5	Use files for data storage.
68185	**Engineering Chemistry Lab**	CO1	Estimate rate constants of reactions from concentration of reactants/products as a function of time.
		CO2	Measure molecular/system properties such as surface tension, viscosity, conductance of solutions, redox potentials, absorbance.
		CO3	Understand the concepts of distribution and adsorption phenomena.
		CO4	Synthesize a small drug molecule.
		CO5	Develop analytical skills and learn how to analyze and present results of an experiment.
65181	**Data Structures through C Lab**	CO1	Familiarity of programming environment in Linux operating system and to translate given algorithms to a working and correct program.
		CO2	Ability to interpret syntax errors as reported by the compilers and to be able to identify and correct logical errors encountered at run time using debuggers like GDB.
		CO3	Ability to write iterative as well as recursive programs.
		CO4	Ability to represent data in arrays, pointers, strings and structures and manipulate them through a program and use them in defining self-referential structures or structures or designing a user defined data type.
		CO5	Ability to implement file processing functions and be able to store, retrieve and process data in text and binary formats.
67182	**IT Workshop**	CO1	Identify the peripherals of PC, assemble, and disassemble PC components.

		CO2	Install the system software MS windows, Linux and dual operating system, required device drivers.
		CO3	Work with productivity tools for word processing, spread sheet and presentations along with designing basic web pages.
		CO4	Understand the main features of the SCILAB program development environment to enable their usage in high learning.
		CO5	Interpret and visualize simple mathematical functions and operations using plots or display**.**

II Year I Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
65201	**Object Oriented Programming Through Java**	CO1	Design and implement object oriented concepts like encapsulation, abstraction and data hiding using programming constructs offered by java language.
		CO2	Realize the power of inheritance, interfaces and packages.
		CO3	Understand and demonstrate the concepts of exception handling and java io streams.
		CO4	Demonstrate knowledge and understanding of multi threading, annotations and applets in java.
		CO5	Design and develop java applications using AWT and Swings for providing solutions to real world problems
65202	**Discrete Structures and Graph Theory**	CO1	Apply formal logic proofs and/or informal, but rigorous, logical reasoning to evolve theoretical proofs to real problems, such as predicting the behavior of software or solving problems such as puzzles.
		CO2	Apply the logical notations to define and reason about fundamental mathematical concepts such as sets, relations, functions, and integers.
		CO3	Understand and appreciate simple proofs of problems result in group theory.
		CO4	Apply the concept of permutations and combinations to problem solving.
		CO5	Demonstrate knowledge of fundamental concepts in graph theory.
67252	**Operating Systems**	CO1	Acquire basic knowledge about different functions, structures and design features of contemporary operating systems, to aid in applying and solving complex engineering problems.
		CO2	Analyze and formulate the problems, design the solutions, implement and demonstrate different

			process scheduling, synchronization, and deadlock related algorithms.
		CO3	Design algorithmic experiments, analyze and interpret the data for different memory management techniques and their implementations.
		CO4	Differentiate and Demonstrate secondary storage structures, file systems, directory structures and their implementations. Select and apply appropriate disk scheduling algorithms to predict the performance.
		CO5	Understand the impact of the different protection and security threats in the context of societal, safety, ethical, legal issues and design solutions and mechanisms including responsibilities to overcome these threats.
68203	**Mathematical and Statistical Foundations**	CO1	Apply the number theory concepts to cryptography domain.
		CO2	Establish relationship between variables using correlation and regression.
		CO3	Find out the appropriate distribution underlying given data.
		CO4	Apply the concepts of fundamental sampling distributions to some case studies.
		CO5	Find estimates of parameters and test hypothesis about parameters.
65205	**Digital Logic and Computer Organization**	CO1	Master different number systems, representation of floating-point number and realize the binary operations of Boolean algebra using logic gates.
		CO2	Solve gate-level minimization problems using K-map.
		CO3	Describe the functional units of the processor, addressing modes, instruction formats, and register transfer language.
		CO4	Design of control unit using micro programming and CPU organization.
		CO5	Explain the concepts of various types of memory in a digital system.
65234	**Object Oriented Programming through Java Lab**	CO1	Implement object oriented concepts like encapsulation, data hiding and abstraction using programming constructs offered by java language.
		CO2	Develop java programs to realize the power of inheritance, interfaces and packages.
		CO3	Develop java programs to demonstrate the concepts of exception handling and I/O streams.
		CO4	Implement java applications using multithreading mechanism.
		CO5	Use graphical user interfaces to create Applets & Swing applications for providing solutions to real world problems.

65235	**Mathematics & Statistics Lab using Python**	CO1	Apply the number theory concepts as needed.
		CO2	Apply the concepts of linear regression and correlation.
		CO3	Apply the concepts of probability distributions.
		CO4	Apply the concepts of sampling distributions.
		CO5	Perform hypothesis testing and estimations.
65236	**Operating Systems and System Administration Lab**	CO1	Install and configure Linux Operating System.
		CO2	Demonstrate the use of Shell commands for handling files.
		CO3	Write programs using I/O system calls for implementing file operations.
		CO4	Simulate and implement operating system concepts such as scheduling, deadlock management, file management and memory management.
		CO5	Implement concurrent programs using process and thread API and establish communication among them.
68232	**Verbal Ability Lab**	CO1	Students will be trained to become proficient in word formation, spellings and vocabulary.
		CO2	Students will develop linguistic competence through appropriate use of Idioms and Phrasal verbs.
		CO3	Students will develop verbal reasoning through verbal analogy.
		CO4	Students will be trained to identify the common errors in English and write grammatically correct sentences.
		CO5	Students will develop professional writing skills through business letters.

II Year II Semester			
Course Code	Course Title	CO Number	Course Outcome Statement
65251	**Advanced Data Structures through Java**	CO1	Realize the power of generics in Java.
		CO2	Understand Java's collection class hierarchy and implement stacks, queues, dictionaries and trees using them.
		CO3	Implement dictionaries using linear lists and hashing, and compare their performance.
		CO4	Implement dictionaries using various height balanced trees and also analyze the advantages and disadvantages of height balanced trees.
		CO5	Evaluate various pattern and word matching algorithms in terms of their complexity and efficiency.
65255	**Computer Algorithms Design and Analysis**	CO1	Analyze worst case running times using asymptotic analysis of algorithms

		CO2	Describe the divide and conquer paradigm and synthesize divide and conquer algorithms.
		CO3	Define optimization problems and solve them through various greedy policies
		CO4	Describe the dynamic programming paradigm and synthesize dynamic programming algorithms, and analyze them.
		CO5	Reduce size of search space of the optimization problems by applying backtracking and branch and bound tools. Appreciate the non-deterministic modeling of algorithms
65256	**Computer Architecture and Micro Controllers**	CO1	Describe the fundamental concepts of parallel processing and multiprocessors.
		CO2	Discuss the problems involving in interfacing peripheral devices with a processor and study serial communication.
		CO3	Explore the internal architecture of 8051 and to create ready to run programs using 8051 assemblers.
		CO4	Explain the importance of Timers/counters in a micro-controller and learn how to program them.
		CO5	Describe the serial communication feature of 8051 and learn how to write interrupt handler programs.
65252	**Data Base Management Systems**	CO1	Demonstrate an understanding of database management system components and features. Design ER represent real-world database application scenarios.
		CO2	Demonstrate mathematical approach towards querying database using relational algebra and relational calculus and implement using SQL.
		CO3	Convert ER model to relational model and design proper relational database while eliminating anomalies.
		CO4	Demonstrate the role of transaction management and concurrency control protocols
		CO5	Demonstrate an understanding of query optimization techniques and apply PL/SQL concepts for database manipulations with constraints.
65258	**Python for Data Science**	CO1	Gain expertise in data handling, and cleaning using python packages.
		CO2	Apply all the functional features of the NumPy to effectively manipulate the data present in arrays.
		CO3	Gain insights into descriptive statistics of categorical and time series data by applying the constructs of Pandas data structure.
		CO4	Master the data interpretation using various exploratory models and gain insights.

		CO5	Build regression models b utilizing various basis functions and evolve prediction mechanisms.
65281	**Advanced Data Structures through Java Lab**	CO1	Implement stacks and queues using user defined generic classes.
		CO2	Implement hashing, sets, stacks and queues using collection classes in java.util package.
		CO3	Implement dictionaries using various data structures like sorted list, and hashing.
		CO4	Implement dictionaries using various height balanced trees and also analyze the advantages and disadvantages of height balanced trees.
		CO5	Implement Pattern Matching Algorithms.
65283	**Database Management Systems Lab**	CO1	Analyze database requirements and determine the entities involved in the system and their relationship to each other.
		CO2	Design E-R Model to represent database application scenarios.
		CO3	Convert/transform the E-R Model to relational tables, populate relational database and formulate SQL queries on data.
		CO4	Improve the database design by normalization.
		CO5	Implement PL/SQL procedures, function, triggers and cursors
65281	**Python for Data Science and DAA Lab**	CO1	Apply basic data handling tools.
		CO2	Gain knowledge of data processing python packages.
		CO3	Apply pandas data frames and visualization to data sets.
		CO4	Solve sorting and searching problems using divide and conquer model.
		CO5	Explore large search spaces using dynamic programming and greedy paradigms.
68283/ 68231	**Reasoning and Data Interpretation Lab**	CO1	Understand the concepts of statement-argument, assumption and course of action and use reasoning as a tool to match statements with argument etc.
		CO2	Look at the data and find link and patterns, link data with conclusions and study data logically
		CO3	Study problem situations and use reasoning as a tool to find solutions.
		CO4	Nurture the ability to use reasoning as a skill in real time problem solving.
		CO5	Analyze and infer the data with respect to trend and case based.
68282	**Gender Sensitization**	CO1	Students will have developed a better understanding of important issues related to gender in contemporary India.
		CO2	Students will be sensitized to basic dimensions of the biological, sociological, psychological and legal aspects of gender, This will be achieved through

			discussion of materials derived from research, facts, everyday life, literature and film.
		CO3	Students will attain a finer grasp of how gender discrimination works in our society and how to counter it.
		CO4	Students will acquire insight into the gendered division of labor and is relation to politics and economics.
		CO5	All the students and professional will be better equipped to work and live together as equals.
		CO6	Students will develop a sense of appreciation of marginalized genders in all walks of life.
		CO7	Through providing accounts of studies and movements as well as the new laws that provide protection and relief to women, the textbook will empower students to understand and respond to gender violence.

III Year I Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
65301	**Computer Networks and Security**	CO1	Appreciate communication reference models along with PDUs and design a physical network based on the topological specifications using various communication media.
		CO2	Design solutions for logical link layer transmission and control errors; control the medium access patterns through an established methodology.
		CO3	Solve the optimal routing problems for static and dynamic networks and realizing various QoS parameters.
		CO4	Demonstrate the significance of various applications using TCP and UDP protocol.
		CO5	Implement CIA security mechanisms, and Network Security protocols.
65308	**Software Engineering for Data Science**	CO1	Understand underlying concepts and principles of software engineering for identifying the requirements and to design a software solution to the problem.
		CO2	Understand the concepts and principles of software engineering for testing the implementation to deliver quality software.
		CO3	Understand the concepts on data systems to design software applications.
		CO4	Understand problems due to data replication in distributed systems and how to design software in such situations.

		CO5	Understand issues arising due to data partitioning in distributed systems and to design software applications to overcome those issues.
67305	**Web Programming**	CO1	Build a custom website with HTML, CSS, Bootstrap and JavaScript.
		CO2	Demonstrate advanced features of JavaScript and learn about JDBC.
		CO3	Develop server side implementation using Java technologies like servlets.
		CO4	Develop the server side implementation using NodeJS and deploy the same.
		CO5	Design a single page application using React.
68304	**Management Science**	CO1	Make business decision for effective business administration.
		CO2	Identify business strategies for effective and efficient utilization of resources.
		CO3	Design the organization which can effectively work in dynamic business environment.
		CO4	Become better leaders and be self-motivated.
		CO5	Develop and implement the control systems in the organization.
65309	**Automata and Compiler Design**	CO1	Employ finite state machine to solve problems in computing and classify machines by their power to recognize languages.
		CO2	Understand the basic concept of compiler design, and it different phases which will be helpful to construct new tools like LEX, YACC etc.
		CO3	Ability to implement semantic rules into a parse that performs attribution while parsing and apply error detection and correction methods.
		CO4	Apply code optimization techniques to improve the space and time complexity of programs while programming.
		CO5	Ability to design a compiler for a concise programming language.
65333	**Web Programming Lab**	CO1	Build a custom website with HTML, CSS, Bootstrap and JavaScript.
		CO2	Demonstrate advanced features of JavaScript and learn about JDBC.
		CO3	Develop server side implementation using Java technologies like servlets.
		CO4	Develop the server side implementation using NodeJS and deploy the same.
		CO5	Design a single page application using React.
67333	**Mobile Application and Development Lab**	CO1	Compare and contrast the language fundamentals of Kotlin.
		CO2	Use Android layouts and connect views with data through data binding.

		CO3	Illustrate the Android activity life cycle mechanism and application architecture.
		CO4	Develop the navigation patterns and display collections of data sing recycler view.
		CO5	Collaborate data persistence in the application, and use APIs to connect to, store, and retrieve data.
65335	**Computer Networks and CASE Tools Lab**	CO1	Articulate modelling and simulation of computer network.
		CO2	Design and implementing basic trouble shooting protocols.
		CO3	Understand and learn CASE tools for software design.
		CO4	Implement open source CASE tools for designing software solutions.
		CO5	Use GitHub and Jenkins Tools for continuous delivery and integration of software solutions.
65332	**Effective Technical Communication Lab**	CO1	Attain proficiency in features of Technical Communication
		CO2	Develop expertise in reading skills.
		CO3	Use English language appropriately to write effective reports, emails, notes and summaries.
		CO4	Become proficient in Analytical and Critical Thinking Skills.
		CO5	Be empowered to use English language effectively in Technical Communication.
68302	**Universal Human Values**	CO1	Understand the significance of values, distinguish between values and skills.
		CO2	Apply the concept of happiness and prosperity to set the goals in life.
		CO3	Evaluate the current scenario in the society, in a right manner.
		CO4	Distinguish between the needs of the self and body through principles of co-existence.
		CO5	Understand the value of harmonious relationship based on trust, respect and other naturally acceptable feelings in human-human relationships.
		CO6	Understand the harmony in nature and existence, and work out their mutually fulfilling participation in the nature.
65409	**Cyber Security**	CO1	Understand the fundamentals of cyber security, including types of cyber threats, attack models, vulnerabilities, and basic defense mechanisms
		CO2	Explain the importance of cyber laws, policies, and forensics tools used for tracking and prosecuting cybercrimes.
		CO3	Analyze the risks and security challenges associated with mobile and wireless computing and propose appropriate solutions.

		CO4	Evaluate the impact of cybercrime on organizations, and assess measures for mitigating security risks and ensuring data protection.
		CO5	Examine privacy-related concerns, ethical issues, and real-world cybercrime cases to derive practical insights and awareness.

III Year II Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
65366	**Data Mining and Machine Learning**	CO1	Understand data kinds, data preprocessing.
		CO2	Describe classical machine learning approach.
		CO3	Apply classification models and tune their accuracy.
		CO4	Apply various clustering methods.
		CO5	Explain reinforcement Learning.
65367	**Data Acquisition and Storage Technologies (DAST)**	CO1	Know the basic concepts of sensors and its protocols for logical design of IoT.
		CO2	Realize the connectivity technologies, hardware platforms and its protocols.
		CO3	Implement IoT with RasberryPi and IoT applications.
		CO4	Understand storing the sensor data in sensor clouds.
		CO5	Design IoT applications in different domains i.e., fog computing, WSN.
65368	**Big Data Analytics and Tools**	CO1	Identify big data problems and understand usage of various tools in Hadoop Ecosystem.
		CO2	Articulate working of HDFS and YARN.
		CO3	Write programs using MapReduce.
		CO4	Use Flume and Sqoop for Data acquis ion and develop applications using NoSQL MongoDB
		CO5	Develop applications using Hive and Apache Pig.
68352	**Managerial Economics and Financial Analysis**	CO1	Familiarize with the fundamentals of Economics such as Demand function, Law of demand, Elasticity of demand and Forecasting methods etc.
		CO2	Evaluate Economies of Scale and the Breakeven point of the business activity.
		CO3	Understand the different Market structures and Macro concepts.
		CO4	Able to understand the accounting system and preparation of Final accounts.
		CO5	Analyze Accounting statements like Income statement and balance sheet to understand financial performance of the business.
65364	**Distributed Systems (PE-I)**	CO1	Elucidate the foundations and issues of distributed systems

		CO2	Understand the various synchronization issues and global state for distributed systems.
		CO3	Understand the Mutual Exclusion and Deadlock detection algorithms in distributed systems
		CO4	Describe the agreement protocols and fault tolerance mechanisms in distributed systems.
		CO5	Describe the features of peer-to-peer and distributed shared memory systems
65355	**Computer Graphics (PE-I)**	CO1	Know the application areas of computer graphics, overview of graphics systems and output primitives.
		CO2	Understand OpenGL as a cross platform application programming interface (API)
		CO3	Apply 2D geometric transforms, 2D viewing using Open Gl transformation matrices.
		CO4	Create programs using OpenGL APIs for rendering 2D and 3D vector graphics.
		CO5	Use visible surface detection methods and computer animation.
65369	**Artificial Neural Networks (PE-I)**	CO1	Understand the similarity of Biological networks and Neural networks and then Perform the training of neural networks using various learning rules.
		CO2	Design Feed-forward neural networks for problems in AI.
		CO3	Understanding the concepts of forward and backward propagations.
		CO4	Design Neural networks based on Understand Self-Organizing maps for unsupervised learning.
		CO5	Understand concept of Neuro-dynamics and build RNN and Hopfield models.
65357	**Agile Methodologies and DevOps (PE-I)**	CO1	Understand the agile principles, practices.
		CO2	Apply roles of Scrum Master and Perform Scrum Release Planning and Scrum Sprint Planning.
		CO3	Apply Scrum with multiple, or distributed, project teams
		CO4	Analyze Need for continuous Integration and Appreciate Continuous Deployment in Industrial Scenario.
		CO5	Implement GitHub and Jenkins for Configuration Management and Continuous Integration.
65370	**Cyber Forensics (PE-I)**	CO1	Understand the fundamentals of Computer Forensics
		CO2	Describe the Computer Forensic Evidence Collection and Preservation of Digital Evidence.
		CO3	Analyze Electronic Evidence and Network Forensics.
		CO4	Compare Countermeasures in Information Warfare
		CO5	Classify Advanced Computer Forensics Systems and Surveillance Tools

65386	**Big Data Analytics and Tools Lab**	CO1	Identify big data problems and understand usage of various tools in Hadoop Ecosystem.
		CO2	Use HDFS for Data storage
		CO3	Write programs using MapReduce.
		CO4	Use MongoDB, Sqoop and Flume for Data transfer and analysis.
		CO5	Develop applications using Hive and Pig.
65387	**DAST and Machine Learning Lab**	CO1	Able to learn the Linux commands to implement on Raspberry Pi board
		CO2	Interface the sensors with Raspberry Pi using various communication models.
		CO3	Implement and learn to sore data in the cloud and retrieve data from cloud
		CO4	Analyze and perform some feature engineering, to transform data suitable for the model.
		CO5	Design and implement various machine learning models to fit real world problems.
65388	**Data Handling and Visualization Lab**	CO1	Understand How to import data into any Visualization tool.
		CO2	Understand the concepts of Dimensions and Measures.
		CO3	Develop Programs and understand how to map Visual Layouts and Graphical Properties.
		CO4	Create Dashboard that links multiple visualizations.
		CO5	Use graphical user interfaces to create Frames for providing solutions to real world problems
68382	**Advanced English Communication & Soft Skills Lab**	CO1	Evolve as effective communicators.
		CO2	Emerge as decision makers, time managers and good negotiators.
		CO3	Gain proficiency in resume writing and requisite interview skills
		CO4	Collate ideas and information and organize them relevantly and coherently.
		CO5	Be empowered to use soft skills in the global context.
68351	**Essence of Indian Knowledge Tradition**	CO1	To gain a general idea of the vast vedic literature and their content and to grasp the relevant concepts of the vedas and appreciate its relevance in the modern world.
		CO2	Understand connect up and explain basics of Indian Traditional Knowledge in modern scientific perspective.
		CO3	Understand yoga psychology as both a positive and a normative science and is contribution for a holistic health.
		CO4	Understand the views of our great philosophers to correlate their efforts to achieve unity in diversity.
		CO5	To gain a general idea of the vast vedic literature and their content and to grasp the relevant concepts

			of the vedas and appreciate its relevance in the modern world.
68320/ 65380	**Fundamentals of AI**	CO1	Introduce the concepts of search strategies.
		CO2	Represent the knowledge and reasoning.
		CO3	Describing the reasoning under uncertainty.
		CO4	Discuss the leaning strategies.
		CO5	Gain the knowledge on Expert systems.

VI Year I Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
65425	**Advanced Big Data Analytics**	CO1	Enables the students to do programming in Scala.
		CO2	Articulate working of Apache Spark.
		CO3	Write programs using Apache Spark API.
		CO4	Develop Big Data Analytics applications using Apache Spark SQL, Streams and Data Frames.
		CO5	Apply Apache SPARK Machine Learning Algorithms for Big Data Analytics.
65426	**Advanced Data Mining**	CO1	Perform OLAP operations on Data Warehouse.
		CO2	Write advanced and complex queries to retrieve data from data cube.
		CO3	Identify knowledge in terms of associations and correlations.
		CO4	Group data objects in large data sets in the form clusters.
		CO5	Mine knowledge from different data sets and Detect outliers.
65427	**Network and System Security (PE-II)**	CO1	Appreciate the mandated security headers introduced in IP version 6.
		CO2	Adapt protocols like SSL, TLS, HTTPS during application development.
		CO3	Effectively learn the popular email security services.
		CO4	Understand the working principles behind wireless LAN security.
		CO5	Become conversant with system security and learn different applications of security.
65466/ 65445	**Social Networks Analysis (PE-II)**	CO1	Develop semantic web related applications.
		CO2	Represent knowledge using ontology.
		CO3	Predict human behavior in social web and related communities.
		CO4	Visualize social networks.
		CO5	Examine social networks analysis using case studies.
65428	**Data Lakes (PE-II)**	CO1	Analyze big data and Data Lake Architecture.
		CO2	Articulate the core of the Lambda architecture and how to apply it in an enterprise.

		CO3	Use data acquisition tools such as Apache Sqoop and Apache Flume.
		CO4	Articulate working of Messaging and Processing tools such as Kafka and Flink.
		CO5	Discuss Indexing, Governance and security issues and challenges in Data Lakes.
65417	**Natural Language Processing (PE-II)**	CO1	Understand regular expressions, information retrieval and extraction.
		CO2	Understand parts- of- speech tagger based on HMMs and transformation-based learning.
		CO3	Comprehensive treatment of natural language generation and of machine translation.
		CO4	Understand how a system would be able to communicate with humans via language and.
		CO5	Inference i.e. drawing conclusions based on facts.
65407	**Wireless Networks and Mobile Computing (PE-II)**	CO1	Apply advanced data communication methods and networking protocols for wireless and mobile environment.
		CO2	Utilize and employ application frame works for developing mobile applications including under disconnected and weakly connected environment.
		CO3	Select components and networks for particular application.
		CO4	Understand issues related to client server computing with adaptation, power aware and context aware computing and MANET Protocols.
		CO5	Have a good understanding of how the underlying wireless and mobile communication networks work, their technical features, and what kinds of applications they can support.
65429/ 65304	**Digital Image Processing (PE-III)**	CO1	Perform power law and gamma correction on images and transform color models
		CO2	Design and develop filters for image enhancement in the Spatial and Frequency domains
		CO3	Summarize the procedure for restoring degraded images and apply region-based segmentation algorithms
		CO4	Represent the image features in a program friendly manner and perform pattern matching algorithms
		CO5	Solve pattern recognition and classification problems by applying Bayesian approach
65430	**Operational Research**	CO1	Represent an engineering problem as an appropriate operations research model
		CO2	Collect and analyze the data in an engineering problem using different operations research methods
		CO3	Draw conclusions after analyzing the data and implementing them in the engineering problem.

		CO4	Make decisions using Game theory in a competitive situation.
		CO5	Take decision on when to replace machines using Replacement models
65465/ 65441	**Quantum Computing (PE-III)**	CO1	Demonstrate knowledge on Qubit states and perform various vector operations
		CO2	Master Linear Algebra applications in Quantum Computing
		CO3	Address uncertainties and measures the observables
		CO4	Master the operation on Tensor and describe the statistical state of a Quantum system using Density operators.
		CO5	Distinguish Quantum state from Measurement and apply the theoretical on different conditions.
65442	**Principles of Financial Management for Data Science (PE-III)**	CO1	Identify the importance of financial management and to gain a broader understanding of concepts, principles and terminology in financial management.
		CO2	Develop awareness on fundamental financial decisions by associating the tools and techniques.
		CO3	Understand various tools of Financial Analysis for Decision Making.
		CO4	Gain awareness and importance of various components of financial system
		CO5	Obtain Knowledge on various types of investments and Valuation.
65411	**Web Services and Cloud Computing (PE-III)**	CO1	Acquire the basic knowledge on Service Oriented Architecture and fundamentals of SOAP & WSDL in defining Web Services.
		CO2	Learn about ROA and ReST based Web Services.
		CO3	Understand the basics of Cloud Computing and explore case studies like Amazon Cloud, Google App Engine, and Micosoft Azure.
		CO4	Learn about Virtualization and case studies like Xen-Para Virtualization, VMWare Full Virtualization.
		CO5	Learn about Federation, Presence, Security and Privacy in the Cloud and also know about the challenges faced in the Cloud.
65437	**Advanced Big Data Analytics Lab**	CO1	Write programming using Scala language.
		CO2	Develop applications using Spark Frame and API.
		CO3	Create applications using Spark SQL and experiment spark streaming.
		CO4	Apply Machine Learning Algorithms in Spark for Big data Analytics.
65437	**Advanced Data Mining Lab**	CO1	Load and read CSV files and perform Descriptive and Predictive operations on data sets

		CO2	Implement different classification techniques based on decision tree
		CO3	Handle missing values in the data sets
		CO4	Extract knowledge in terms of frequent item sets and association rules
		CO5	Group data objects into clusters using python libraries
65433	**Industry Oriented Mini Project**	CO1	The graduate will be able to explain the aim, objective and utility of the complex problem to selected audience.
		CO2	The graduate will be able to develop software Project management skills, Problem solving skills and System integration skills.
		CO3	The graduate will be able to work in a team to solve real life problems and maintain professionalism.
65434	**Technical Seminar-I**	CO1	The graduate will be able to comprehend the design and development of imparted knowledge in different aspects of a domain.
		CO2	The graduate will be able to develop presentation skills, build confidence and improve communication skills.
		CO3	The graduate will be able collect ideas through literature survey about new innovations, analyze and present them.
		CO4	The graduate will be able to sharpen his/her personality, intelligence and make themselves aware of knowledge about new hardware and software needs of market

IV Year II Semester			
Course Code	**Course Title**	**CO Number**	**Course Outcome Statement**
65462	**Speech Recognition (PE-IV)**	CO1	Understand the ideas of speech signal analysis and artificial intelligence ideas.
		CO2	Understand a system which increases the robustness of the signal representation.
		CO3	Understand HMMs, is a powerful set tools for providing a statistical model.
		CO4	Understand the Relations between HMMs, Bayesian and Neural Networks.
		CO5	Understand the technology behind Automatic Speech Recognitions (ASR) from theoretical point of view.
65469	**Supply Chain Analytics (PE-IV)**	CO1	Articulate how supply chain works and its operations.
		CO2	Demonstrate supply chain analytics components and benefits in practice

		CO3	Use supply chain analytics to enhance SC strategy and design processes.
		CO4	Apply supply chain analytics to enhance SC Execution processes.
		CO5	Use supply chain analytics to enhance supply chain people processes and deal organizational social issues to use SC Analytics.
65453	**Soft Computing (PE-IV)**	CO1	Understand Soft Computing Techniques such as Neural Networks and Rough Sets
		CO2	Understand and use Associative Memory and Adaptive Resonance Theory.
		CO3	Use Fuzzy Set Theory and develop Fuzzy systems and applications.
		CO4	Understand and apply Genetic Algorithms and Modeling for finding solutions real life problems.
		CO5	Understand Hybrid systems that use combination of soft computing techniques.
67451	**Wireless Ad-hoc and Sensor Networks (PE-IV)**	CO1	Understand basics of MANETs and routing protocols
		CO2	Understand how TCP modified for wireless networks
		CO3	Design of different layers of WSN
		CO4	Understand issues and challenges of security in WSNs
		CO5	Design and implement sensor network protocols in the NesC/TinyOS
65470/65375	**Cloud Security (PE-IV)**	CO1	Demonstrate the growth of Cloud computing, architecture, and different modules of implementation
		CO2	Evaluate the different types of cloud solutions among IaaS, PaaS, SaaS
		CO3	Access the security implementation flow, actions and responsibilities of stake holders
		CO4	Generalize the Data Centre operations, encryption methods and deployment details.
		CO5	Provide recommendations for using and managing the customer's identity and choose the type of virtualization to be used.
65471/ 65361	**Computer Vision and Deep Learning (PE-V)**	CO1	Implement 2D and 3D transformations on images
		CO2	Extract valid features from the images using adaptive statistical tools
		CO3	Solve classification and segmentation problems using CNN based tools
		CO4	Appreciate the mathematical intuitions behind generative models and implement GAN architecture.

		CO5	Implement the next word prediction algorithm using CBOW and LSTM
65423/ 67455	**Block Chain Technologies (PE-V)**	CO1	Understand the structure of a blockchain and why/when it is better than a simple distributed database.
		CO2	Evaluate the setting where a blockchain based structure may be applied, its potential and its limitations.
		CO3	Understand what constitutes a ―smart contract, what are its legal implications and what it can and cannot do, now and in the near future.
		CO4	Attain awareness of the new challenges that exist in monetizing businesses around blockchains and smart contracts.
		CO5	Describe and understand the differences between the most prominent blockchain structures and permissioned blockchain service providers, as well as rising alliances and networks.
67454	**Design Patterns (PE-V)**	CO1	Appreciate the basic concepts of design patterns and able to know how to select and use the design patterns.
		CO2	Identify the design pattern in the existing code and use of creational patterns.
		CO3	Apply and use the structural patterns
		CO4	Identify and use the behavioral patterns
		CO5	Find and catalog patterns in the object-oriented software
65456	**Spatial Informatics(PE-V)**	CO1	Understand spatial concepts, spatial data models and spatial query languages.
		CO2	Implement the spatial operations, spatial queries and query optimization.
		CO3	Design data models with networks. Implement accessing methods.
		CO4	Analyze the spatial data using GIS timeline, and also perform statistical analysis on spatial data.
		CO5	Work on Commercial software or tools and can analyze the commercial trend in spatial information industry.
65472	**Marketing Management for Data Science (PE-V)**	CO1	Analyze the needs of the customer and the dynamic market
		CO2	Segment, target and position a product to increase customer value
		CO3	Determine the pricing strategy to increase the demand for a product
		CO4	Use various techniques for predictive analysis in Retail Sector
		CO5	Evaluate the need for digital evolution in marketing and retail sector specifically

65481	**Technical Seminar-II**	CO1	Comprehend the design and development of imparted knowledge in different aspects of a domain.
		CO2	Develop presentation skills, build confidence and improve communication skills.
		CO3	Collect ideas through literature survey about new innovations, analyze and present them.
		CO4	Sharpen his/her personality, intelligence and make themselves aware of knowledge about new hardware and software needs of market.
65482	**Project Work**	CO1	Define a problem with real-world motivation and can analyze the user requirements.
		CO2	Review the literature related to the problem and identify the gaps in existing methods.
		CO3	Apply problem-solving and technical skills in designing a system to obtain solution.
		CO4	Implement the project and perform results analysis, testing and validation of the solution.
		CO5	Ability to prepare the documentation for the project.